

Mini HIL Protocol Frame Examples

Frame Format: [SOF][CMD/RSP][LEN][DATA...][EOF]
SOF=0xAA, EOF=0x55

Example	Frame	Description
Ping TX	AA F0 00 55	Request firmware version
Pong RX	AA F1 02 01 02 55	Firmware v1.2
GPIO2 ON TX	AA 10 02 02 01 55	GPIO2 ON
GPIO ACK RX	AA 01 01 10 55	ACK for GPIO_SET
Relay1 OFF TX	AA 20 02 01 00 55	Relay1 OFF
Relay ACK RX	AA 01 01 20 55	ACK for RELAY_SET
ADC Start TX	AA 30 03 00 E8 03 55	ADC0 @1000Hz
ADC ACK RX	AA 01 01 30 55	ACK for ADC_START
ADC Data RX	AA 02 03 00 B2 0C 55	ADC0 value=3250
WaveGen TX	AA 40 05 00 F4 01 B8 0B 55	Sine,500Hz,3000mV
Button State RX	AA 04 02 02 01 55	Button2 pressed
GPIO State RX	AA 03 02 04 01 55	GPIO4 HIGH
Error RX	AA FF 02 10 03 55	GPIO_SET failed: Invalid channel

ACK Recommendation

Use ACK frames containing the original command:

AA 01 01 <CMD> 55

Examples:

GPIO ACK: AA 01 01 10 55

Relay ACK: AA 01 01 20 55

ADC ACK: AA 01 01 30 55